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Assignment 5

```sql
CREATE TABLE books(
    bookid INT PRIMARY KEY,
    title VARCHAR(40),
    author VARCHAR(40),
    category VARCHAR(40),
    price DOUBLE
    )

    SELECT * FROM books;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

bookid	title	author	category	price
2	1984	George Orwell	Dystopian	12.5
46	A Farewell to Arms	Ernest Hemingway	Classic	12.75
37	American Gods	Neil Gaiman	Fantasy	16.5
18	Animal Farm	George Orwell	Dystopian	9.99
27	Anna Karenina	Leo Tolstoy	Historical	15.5
13	Brave New World	Aldous Huxley	Dystopian	13.5
28	Catch-22	Joseph Heller	Satire	13.25
11	Crime and Punishment	Fyodor Dostoevsky	Classic	14.5
21	Dracula	Bram Stoker	Horror	11.25
4	Dune	Frank Herbert	Sci-Fi	18
50	East of Eden	John Steinbeck	Historical	16.75
33	Ender's Game	Orson Scott Card	Sci-Fi	14.75
15	Fahrenheit 451	Ray Bradbury	Dystopian	11.99
5	Foundation	Isaac Asimov	Sci-Fi	16.5
22	Frankenstein	Mary Shelley	Horror	10.75
39	Good Omens	Terry Pratchett	Fantasy	14.25
32	Hyperion	Dan Simmons	Sci-Fi	17

```
// set up an HTTP server off port 3000
const express = require("express");
const app = express();
const port = 3000;

//*************************************************************************
*********
//*** set up mysql connections
//*************************************************************************
*********
const pool = require('./mysql');

//*** server waits indefinitely for incoming requests
app.listen(port, function () {
    console.log("NodeJS app listening on port " + port);
});
```

```
//*********************************************************************
*********
//*** receive the get request
//*********************************************************************
*********
app.get("/", async (req, res) => {
    //*** extract the search and format parameters from the query
string
    const category = req.query.category;
    const author = req.query.author;
    const format = req.query.format;

    let sql_query = "SELECT * FROM books";
    const params = [];

    if (category) {
        sql_query += " WHERE category = ?";
        params.push(category);
    } else if (author) {
        sql_query += " WHERE author = ?";
        params.push(author);
    }

    try {
        //*** pool.execute returns [rows, fields]
        const [result, fields] = await pool.execute(sql_query, params);

        if (format === "text") {
            let html_body = "<HTML><HEAD><META
CHARSET='UTF-8'></HEAD>";
            html_body += "<STYLE>body{font-family:arial}</STYLE>";
            html_body += "<BODY><TABLE BORDER=1>";

            //*** print column headings
            html_body += "<TR>";
            for (let i = 0; i < fields.length; i++) {
                html_body +=
(`<TH>${fields[i].name.toUpperCase()}</TH>`);
            }
            html_body += "</TR>";

            //*** print rows of table data
            for (let i = 0; i < result.length; i++) {
```

```
                html_body += (`<TR><TD>${result[i].bookid}</TD>` +
                              `<TD>${result[i].title}</TD>` +
                              `<TD>${result[i].author}</TD>` +
                              `<TD>${result[i].category}</TD>` +
                              `<TD>${result[i].price}</TD></TR>`);
            }

            html_body += "</TABLE></BODY></HTML>";
            res.send(html_body);
        } else {
            //*** JSON is the default when format is missing or
anything else
            res.json(result);
        }

    } catch (err) {
        //*** log the detailed error on the server
        console.error("SQL Error:", err);
        res.status(500).send("<html><body><h1>Database Query
Error</h1><p>An error occurred while fetching
data.</p></body></html>");
    }
});
```



